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Stellenbosch University

Lecture 5

QA344: Quality Assurance
Gehalteversekering

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Learning outcomes for Chapter 5: Don't come to the test without BEING ABLE TO APPLY this!!

- To define a **process** and how it involves combinations of people, machines, tools, techniques, materials, and improvements in a defined series of steps or actions.
- To introduce the concept of **process management**, which involves planning and administering the activities necessary to achieve a high level of performance in value creation and support processes, including their design, control, and improvement.
- To understand that cycle time and the reduction thereof is an important concept in process design, and refers to the time it takes to accomplish one cycle of a process
- To define **value-creation processes** (sometimes called core processes) as those most important to "running the business" and maintaining or achieving a sustainable competitive advantage
- To appreciate the importance of **Support processes** which are those that are most important to an organization's value-creation processes, employees, and daily operations.
- To have a good understanding of leading process management practices related to value creation and support processes
- To be able to do process design that will satisfy internal and external customers
- To understand that in manufacturing, process design usually involves detailed technical analysis of product characteristics,

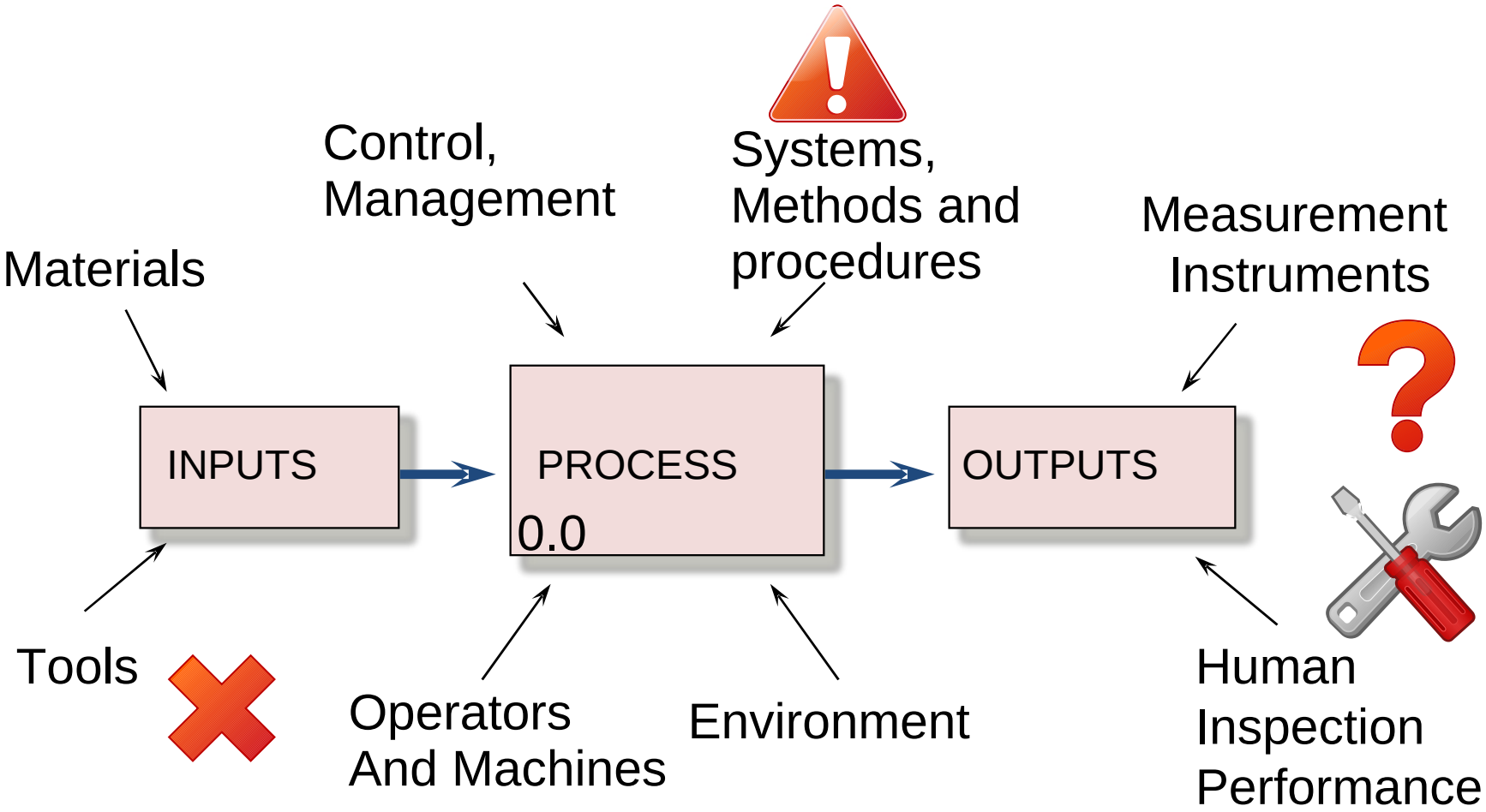
Learning outcomes for Chapter 5: Don't come to the test without BEING ABLE TO APPLY this!!

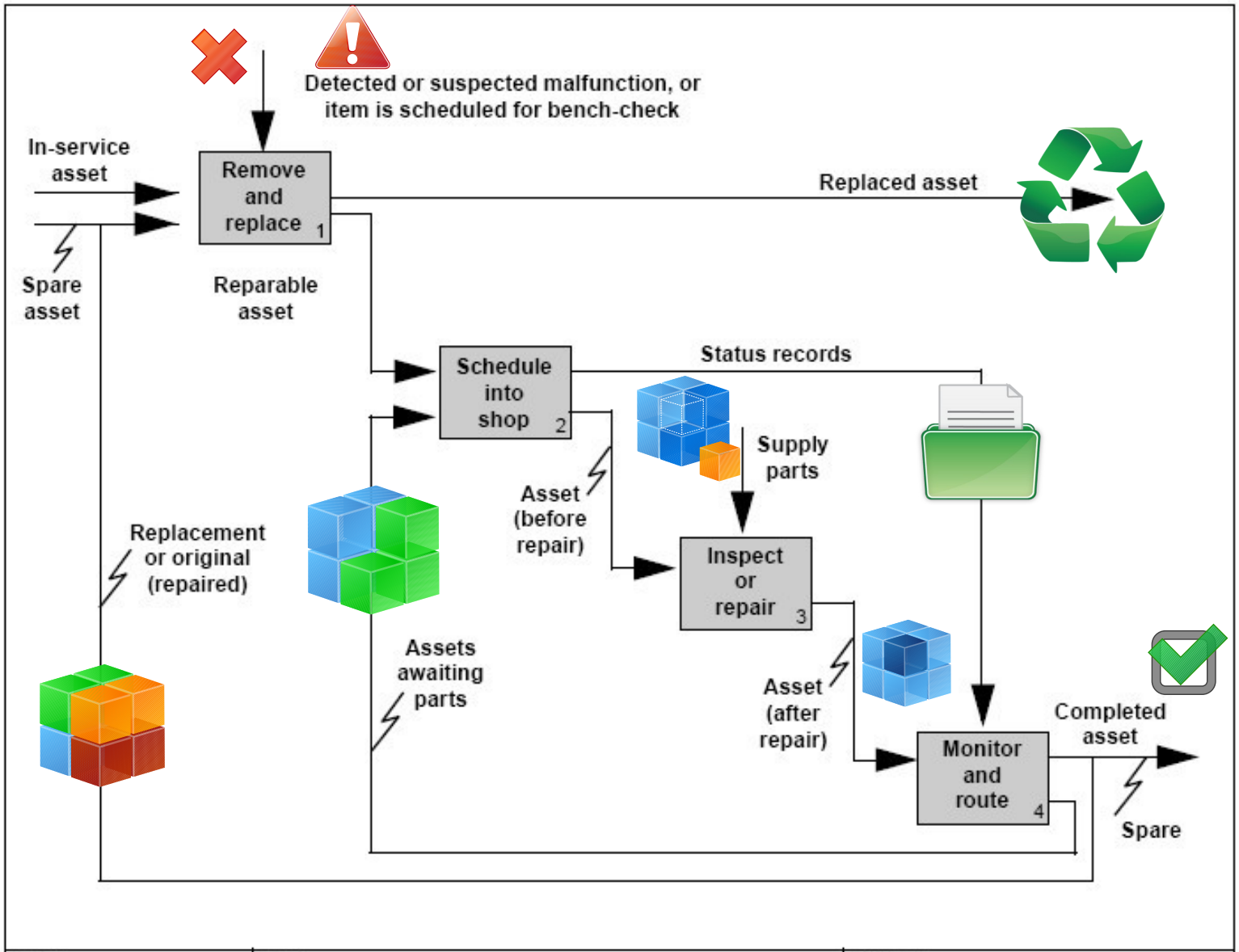
- To design and improve service processes designed around the five key service dimensions of: reliability, assurance, tangibles, empathy, and responsiveness.
- To know that as customer needs and expectations change, organizations must design processes that are increasingly agile. Agility is a term that is commonly used to characterize flexibility and short cycle times.
- To define Poka-yoke (POH-kah YOH-kay) as an approach for mistake-proofing processes using automatic devices or methods to avoid simple human error.
- To learn that control is the activity of ensuring conformance to requirements and taking corrective action when necessary to correct problems and maintain stable performance.
- To understand the four components of a control system: (1) *a standard or goal,* (2) *a means of measuring accomplishment,* (3) *comparison of results with the standard to provide feedback,* and (4) *the ability to make corrections as appropriate* throughout a manufacturing or service process.
- To encourage understanding and development of documented control procedures
- To explore the concept **of continuous improvement** which refers to both incremental changes, which are small and gradual, and breakthrough improvements, which are large and rapid.
- To touch on other approaches to process design and improvement include: 1) breakthrough improvement; 2) benchmarking and 3) reengineering, the fundamental rethinking and radical redesign of business processes

Processes

- A process is a sequence of linked activities that is intended to achieve some result.
- Processes involve combinations of people, machines, tools, techniques, materials, and improvements in a defined series of steps or actions.
- Examples:
  - machining
  - mixing
  - assembly
  - filling orders,
  - approving loans

Source of variation in the production process





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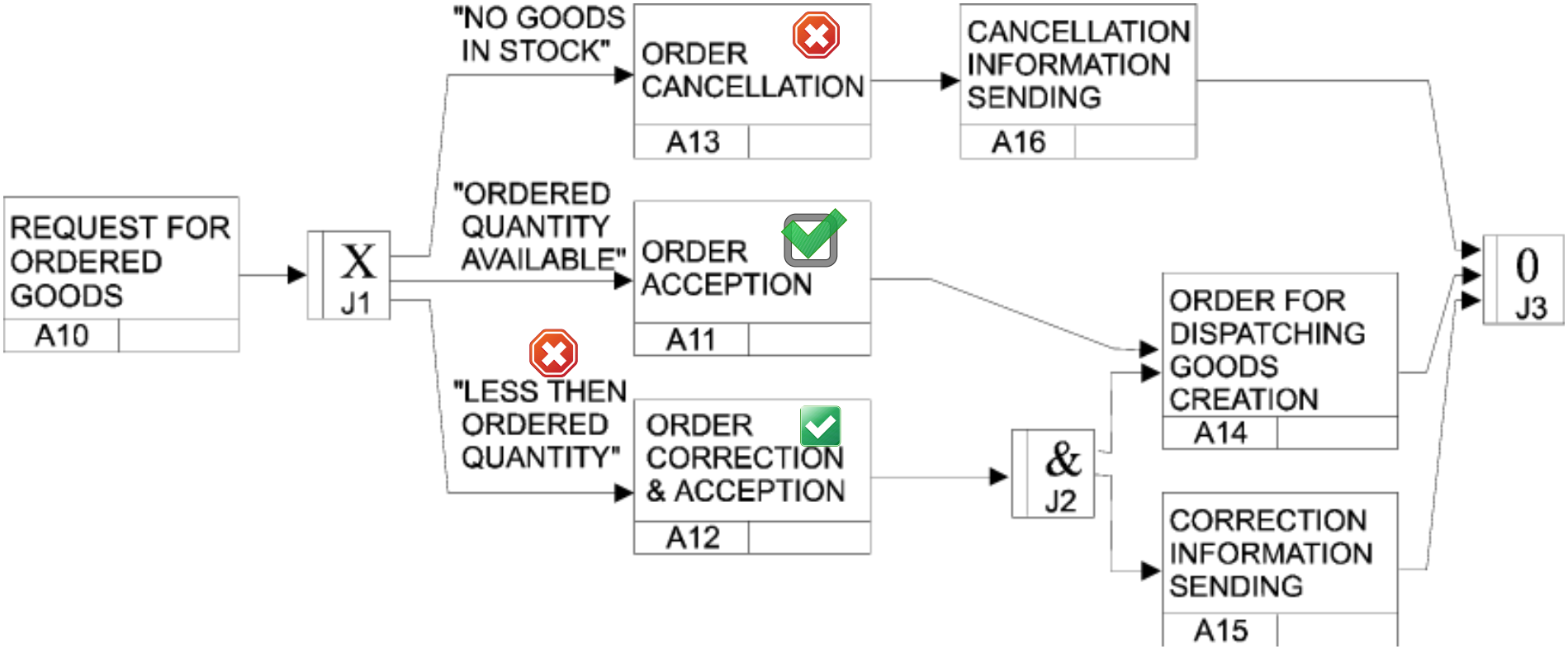
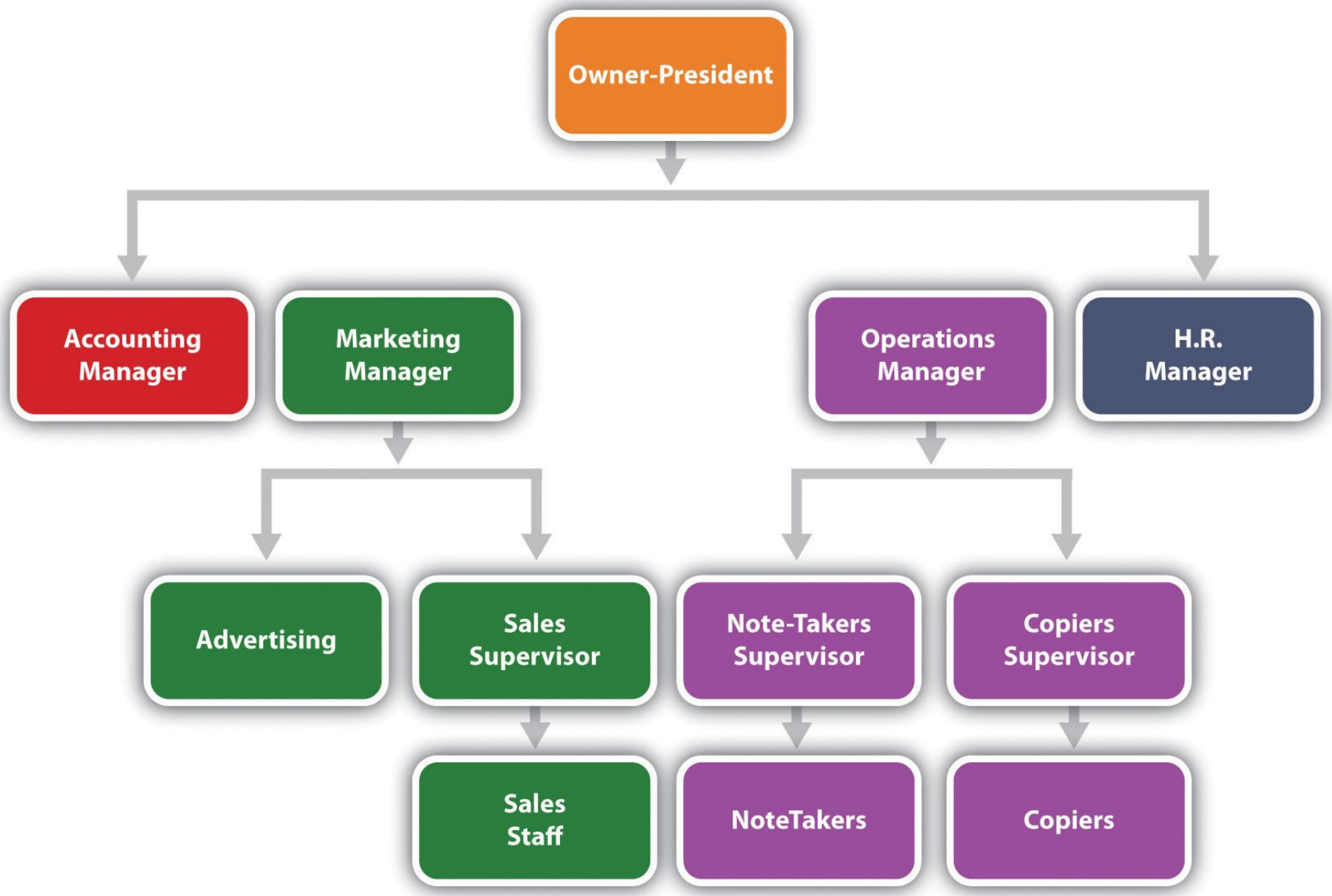
A0F	Maintain Reparable Spares	pg. 4–5

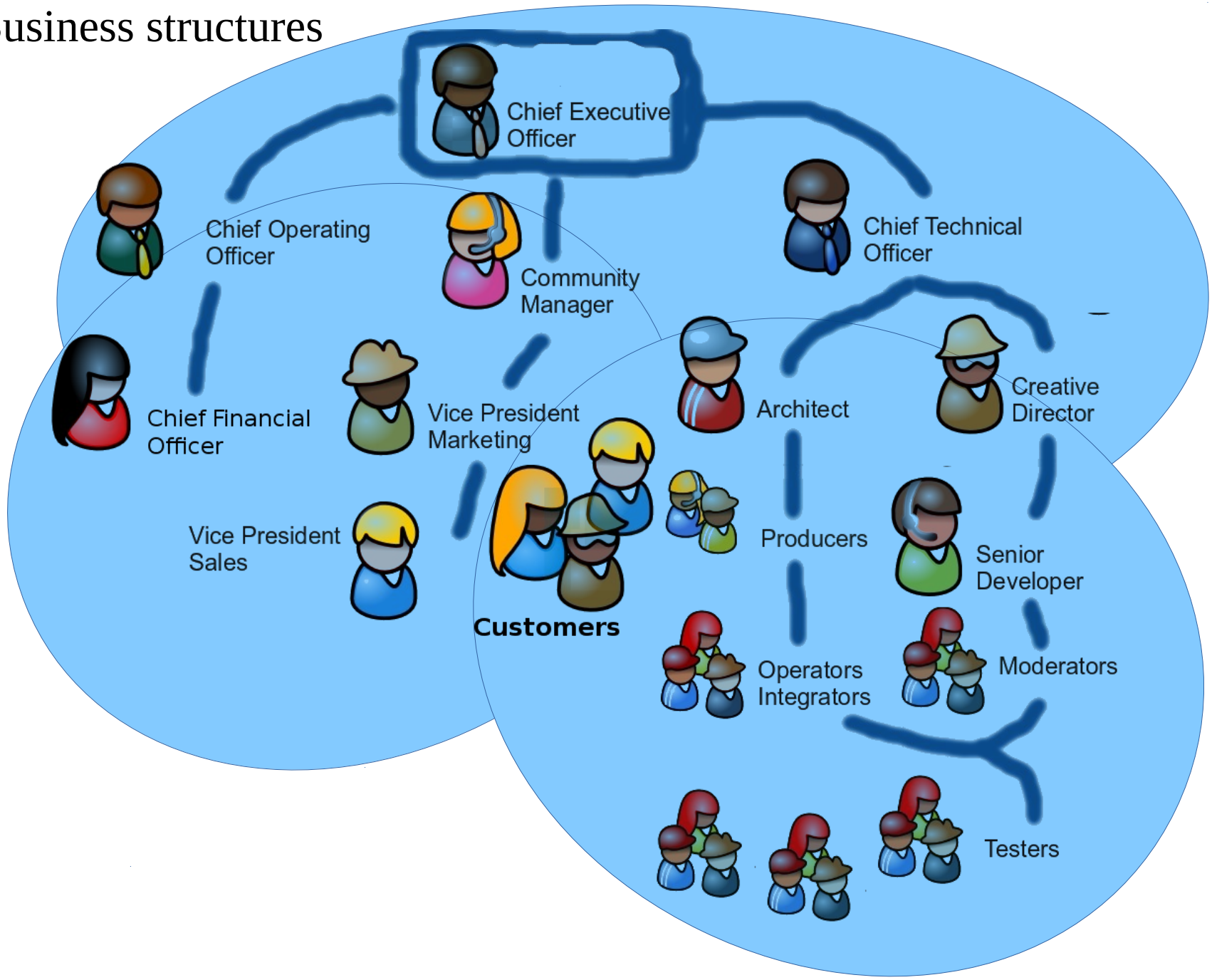
Many standards of describing processes (see tut for details)



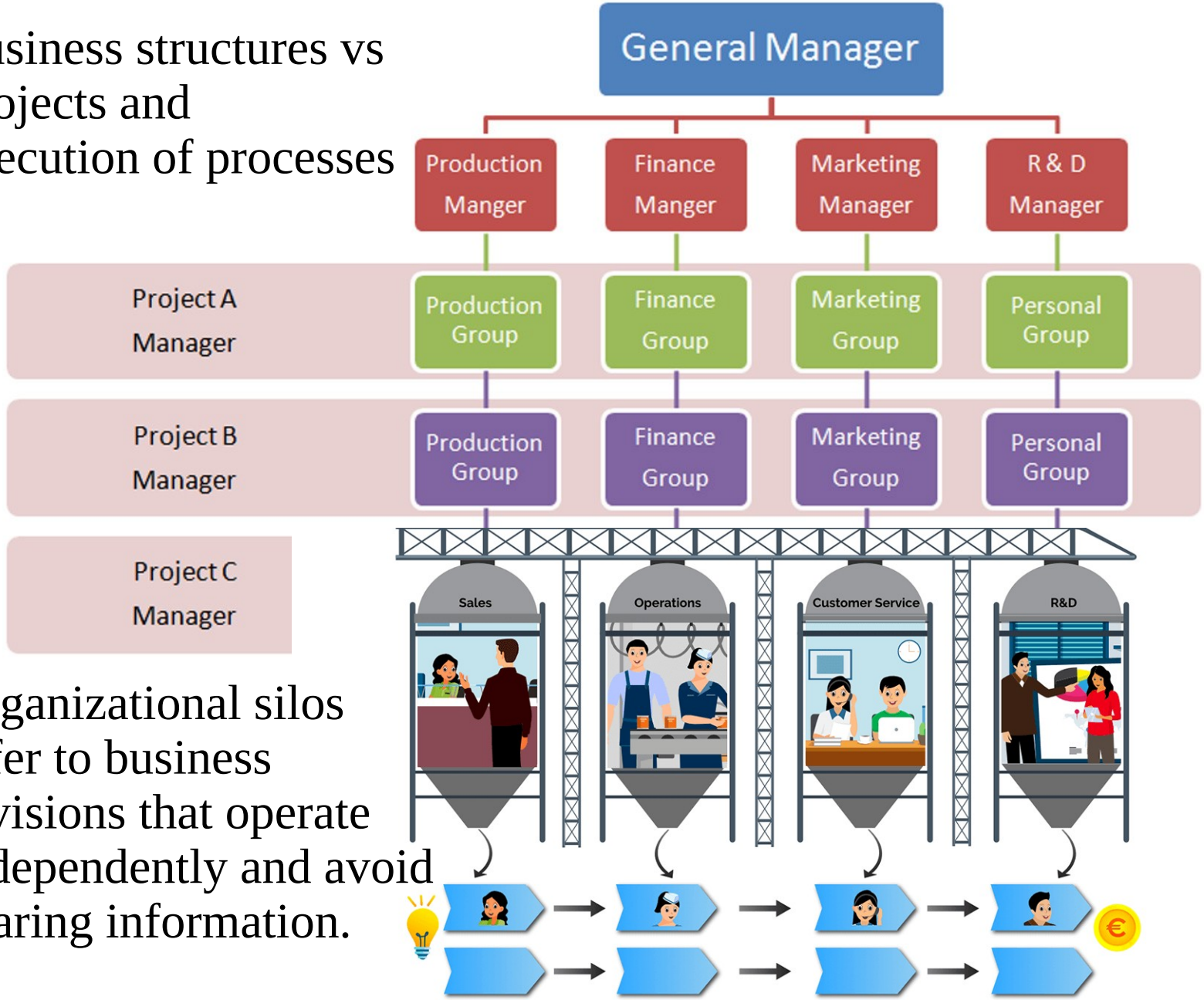
Business structures



Business structures

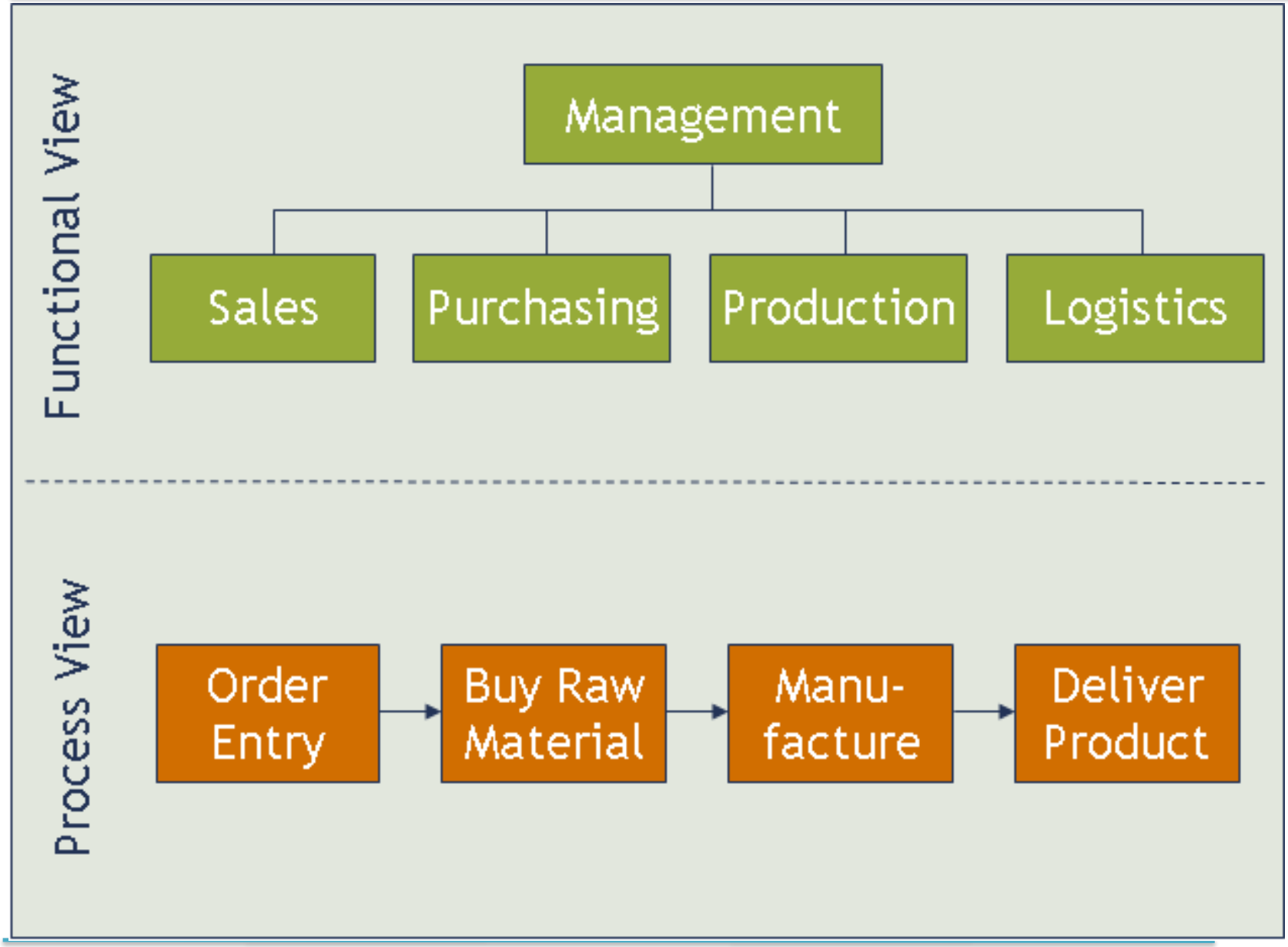


Business structures vs Projects and execution of processes

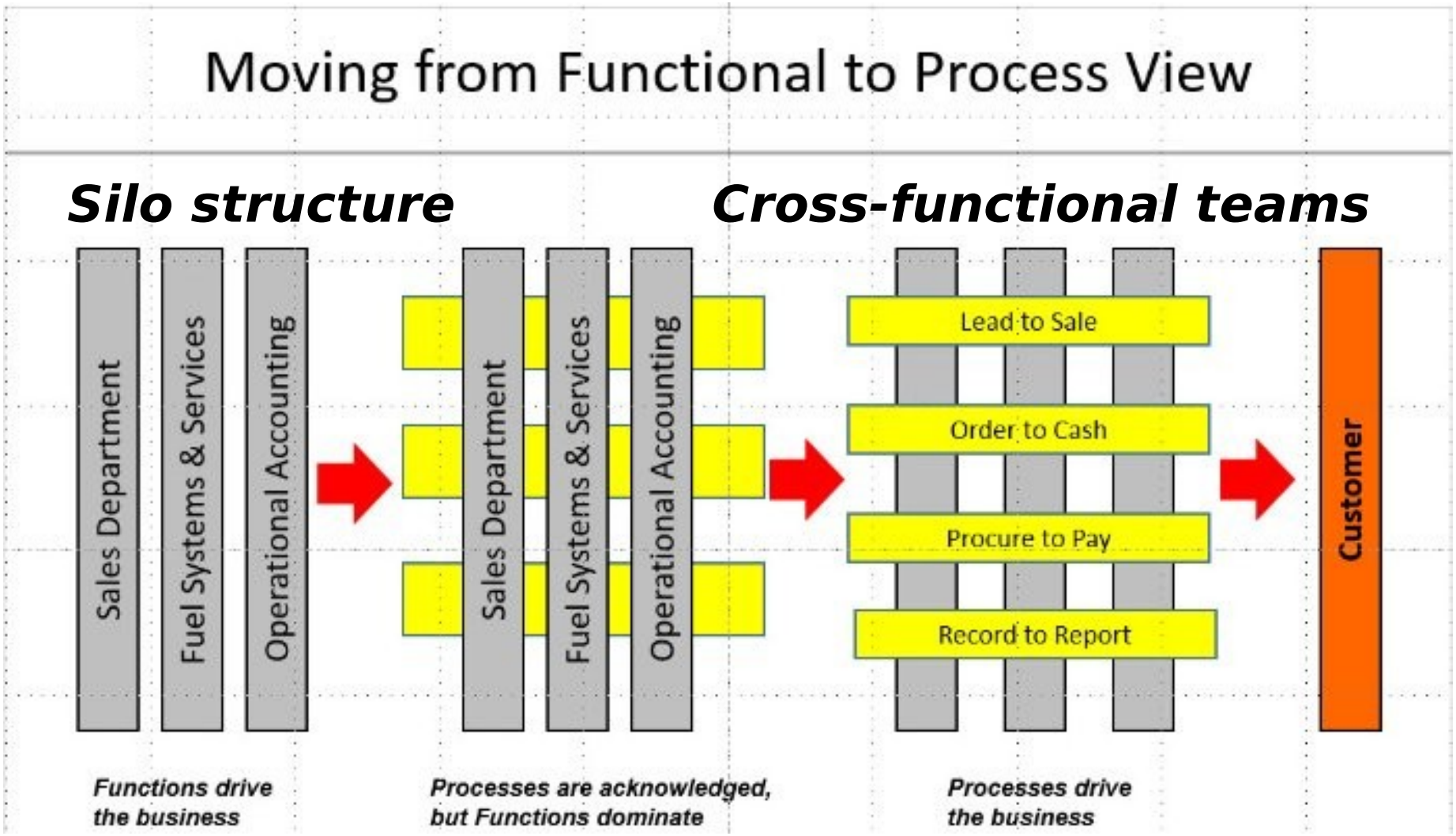


Organizational silos refer to business divisions that operate independently and avoid sharing information.

Process Versus Function



Process Versus Function



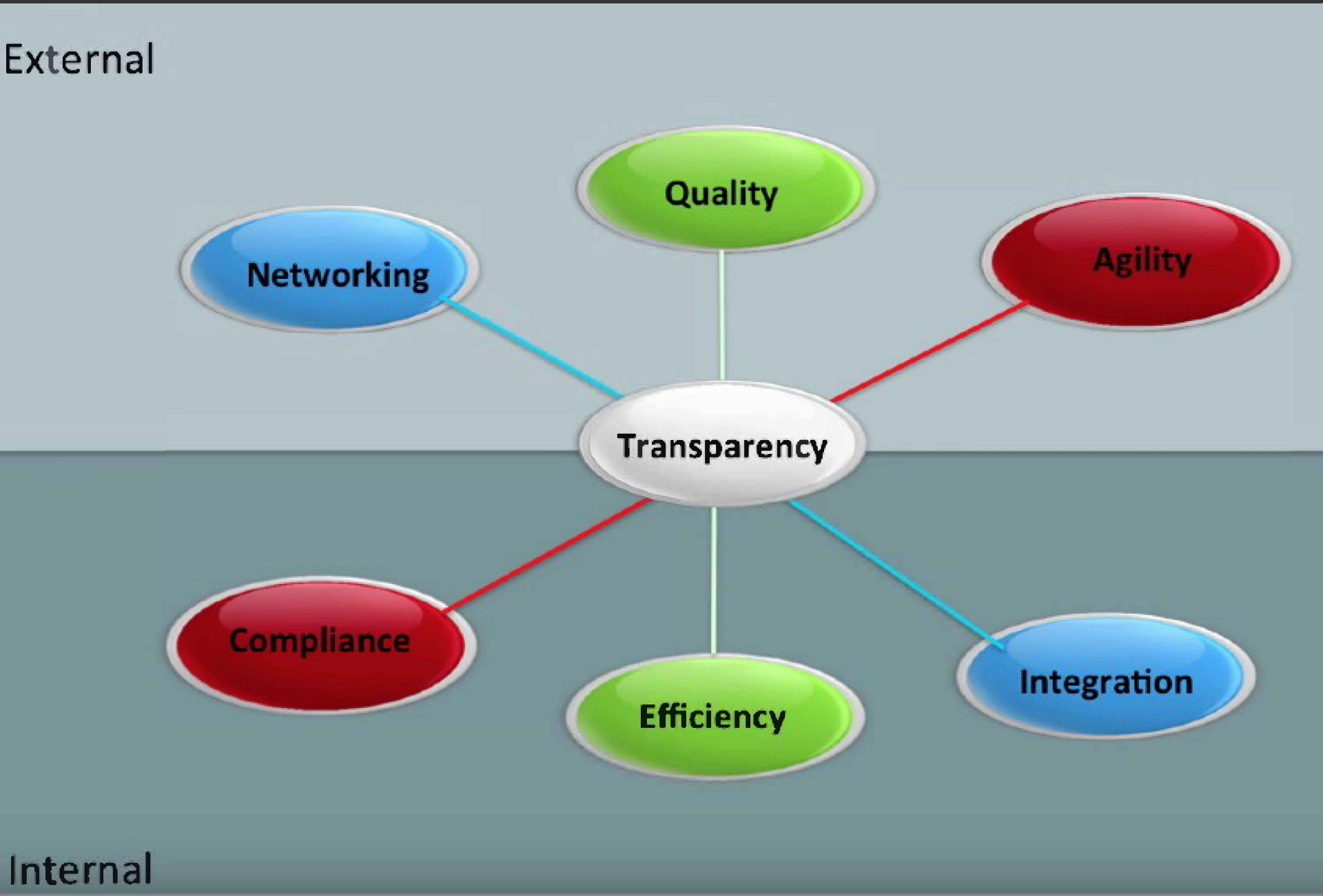
Process-Focused Practices for Performance Excellence

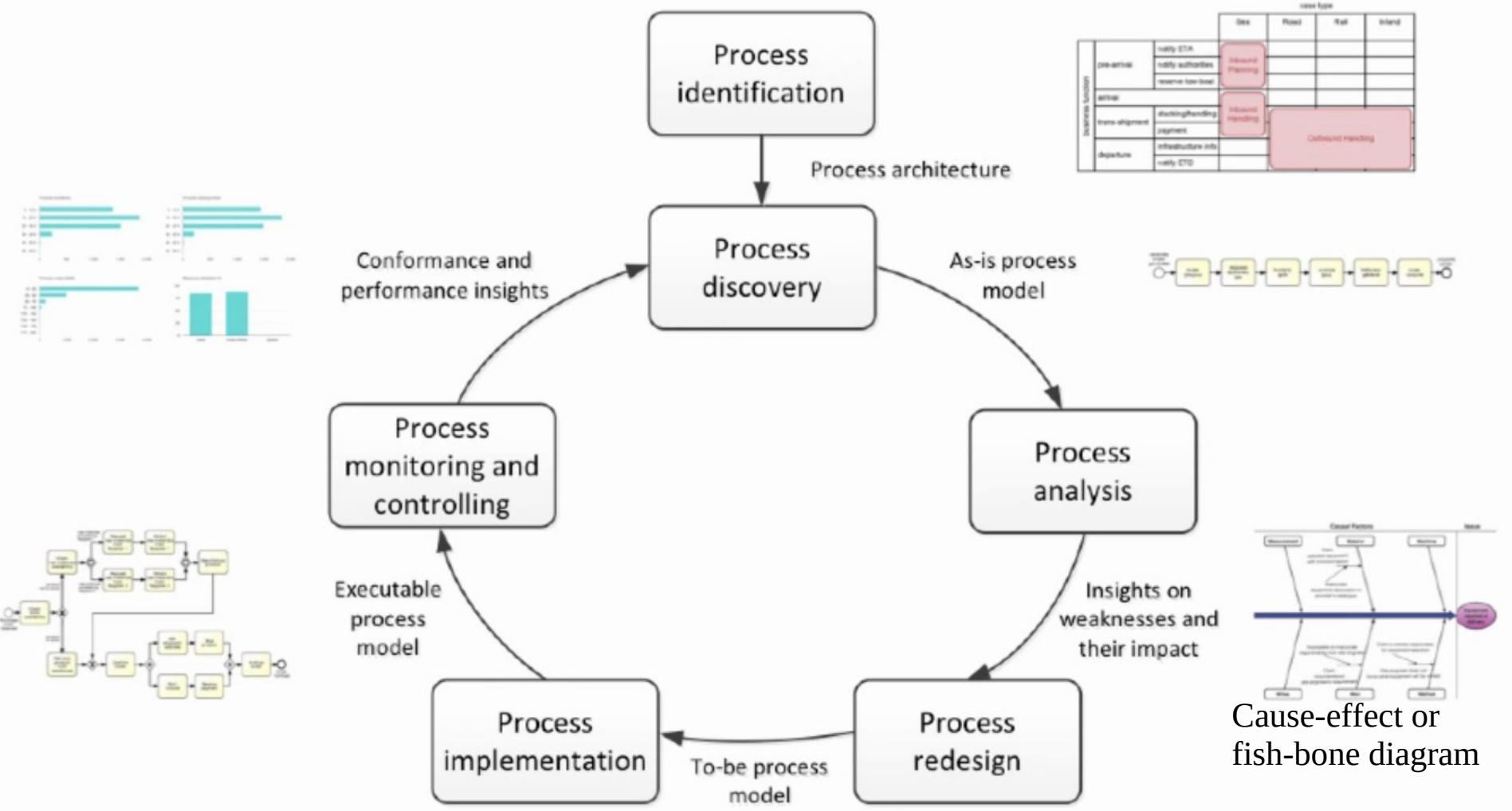
- Practice 1: Identify **vital work processes that relate to core competencies** and deliver customer value, profitability, organizational success, and sustainability.
- Practice 2: Determine **key work process requirements**, incorporating input from customers, suppliers, partners, and collaborators.
- Practice 3: **Design and innovate work processes** to meet all requirements, incorporating new technology, organizational knowledge, cycle time, productivity, cost control, and other efficiency and effectiveness factors.
- Practice 4: **Seek ways to prevent defects, service errors, and rework and minimize costs** associated with inspections, tests, and process or performance audits.

Key Workforce-Focused Practices for Performance Excellence

- Practice 5: **Implement work processes and control their day-to-day operation** to ensure that they meet design requirements, using appropriate performance measures along with customer, supplier, partner, and collaborator input as needed.
- Practice 6: **Improve work processes to achieve better performance**, reduce variability, improve products and services, and keep processes current with business needs and directions, and share improvements with other organizational units and processes to drive organizational learning and innovation.
- Practice 7: Incorporate effective process management practices in the **overall supply chain.**

https://insideevs.com/news/673158/tesla-giga-casting-manufacturing-becomes-mainstream/




Process identification
Process architecture
Process discovery
As-is process model
Process analysis
Insights on weaknesses and their impact
Process redesign
To-be process model
Process implementation
Executable process model
Process monitoring and controlling
Conformance and performance insights
Cause-effect or fish-bone diagram

Process Management: Design, Control and Improve

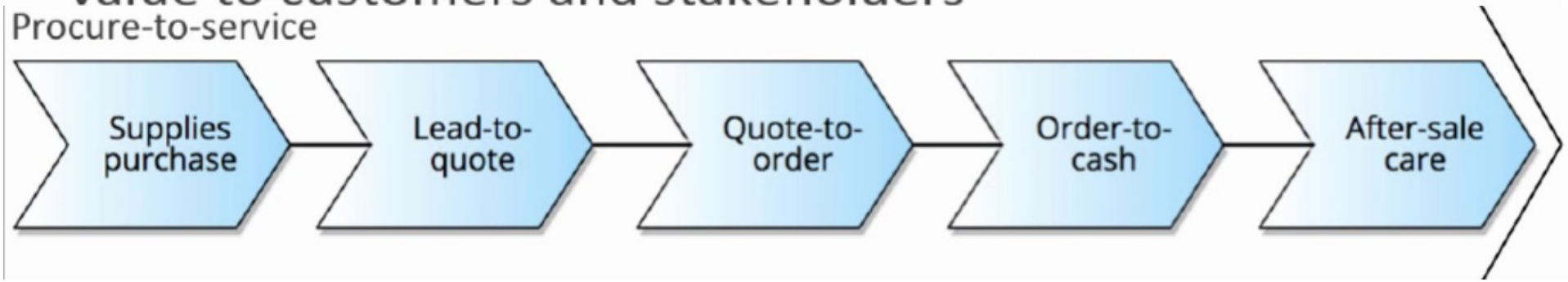
- … *involves planning and administering the activities necessary to achieve a high level of performance in key business processes, and identifying opportunities for improving quality and operational performance, and ultimately, customer satisfaction.*

Process Management Activities

- Design: Ensuring that the inputs to the process, such as materials, technology, work methods, and a trained workforce are adequate, and that the process can achieve its requirements.
- Control: Maintaining consistency in output by assessing performance and taking corrective action when necessary.
- Improvement" Continually seeking to achieve higher levels of performance in the process, such as reduced variation, higher yields, fewer defects and errors, smaller cycle times, and so on.

Horizontal Scoping: Value Chains

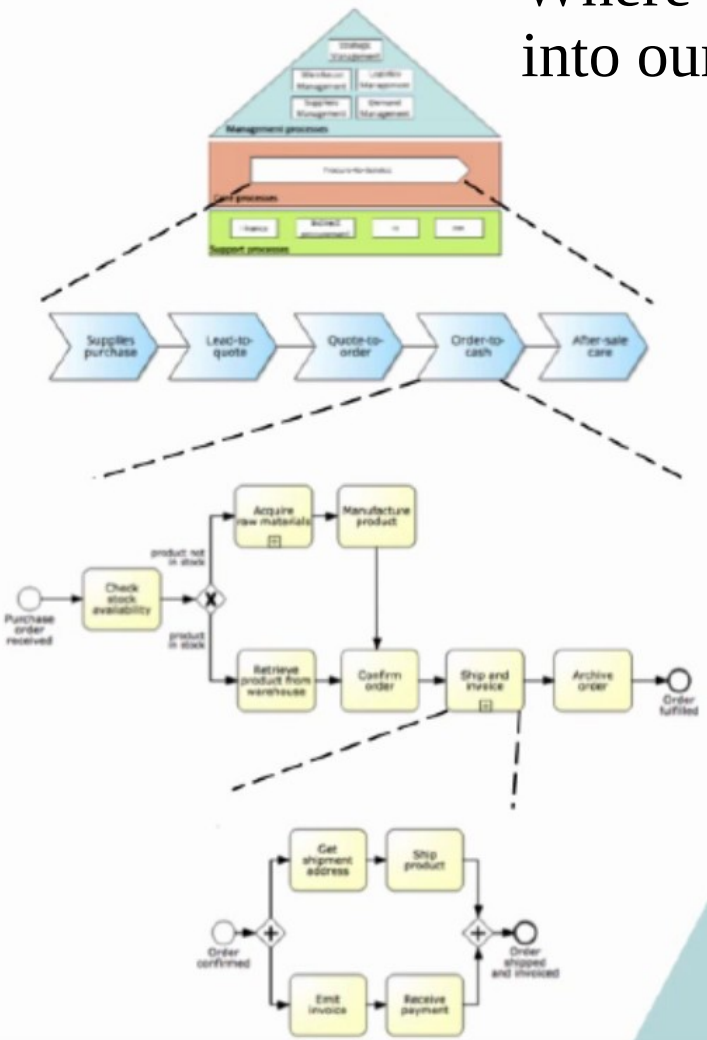
- Chain of *core processes* an organization performs to deliver value to customers and stakeholders

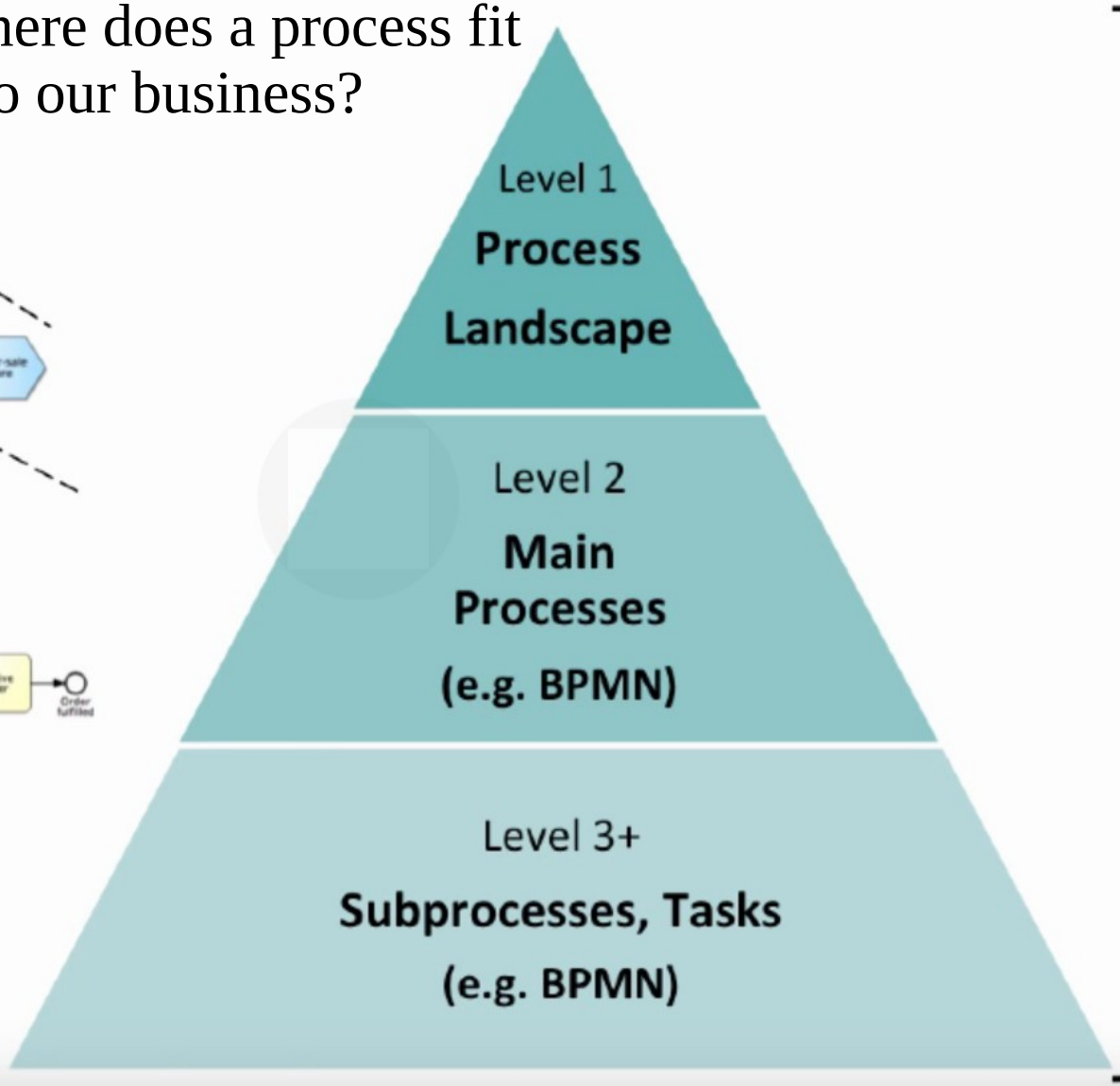


- Value-creation processes – those most important to "running the business"
  - Design processes – activities that develop functional product specifications
  - Production/delivery processes – those that create or deliver products
- Support processes – those most important to an organization's value creation processes, employees, and daily operations

Vertical Scoping: Process Hierarchies

Where does a process fit into our business?





Process Requirements to be of value to business…

- **Value creation process requirements** usually depend on **consumer or external customer needs.**
  - **e.g. fast check-in**
- Support process requirements are driven by internal customer needs and must be aligned with the needs **of key value-creation processes**
  - **E.g. IT that supports check-in**

Some examples (Read and understand how to differentiate value vs support)

	Value	Support
Business	Production design, production, delivery, research and development	Supply chain, maintenance, IT, HR, legal services, billing and accounting, marketing
Health	Delivery design, patient care and treatment, clinical diagnosis	Admissions, food, institutional development, public relations, diagnostic imaging, medical records
Education	Education program, program delivery, student services, curriculum development	Buss and transport, counselling, extra curricular activities, in-service education, administration

Project management – as process steps...

Processes	Description
Phase management	Make sure that you fully understand the "gates", or deliverables that must be completed and approved by the appropriate stakeholders
Planning	High-level planning for the whole project at the start of the project, then do more detailed planning for each phase at the start of each phase
Control	control scope , cost and issues ; and to manage time, risks, and benefits
Team management	Managing the project team.
Communication	Make sure that you're clear about who is responsible for communicating to team members, the project board, the different stakeholders
Procurement	Specialist area – may hire third parties
Integration	Many projects do not stand on their own within an organization – they impact other areas of the business. Make sure that you consider if your project will interface with other projects or functions.

Process Design – what is it?

- The goal of process design is to **develop an efficient process that satisfies both internal and external customer requirements** and is capable of achieving the requisite level of **quality and performance.**
- Process design considerations **include safety, cost, variability, productivity, environmental impact, "green" manufacturing, measurement capability, and maintainability** of equipment.

Process analysis techniques

Inventory, Motion, Over-Processing, Overproduction, Waiting, Transport, and Defects.

- **Lean**
  - Waste analysis
  - Six big losses
  - Bottleneck analysis
- **Six Sigma**
  - Value-added analysis
  - Root-cause analysis
  - Pareto analysis
- **Business Analysis**
  - SWOT
  - Stakeholder analysis
  - Issue Register
- **Operations Mgnt.**
  - Flow Analysis
  - Queuing Analysis
  - Simulation

Unplanned Stops, Planned Stops, Small Stops, Slow Cycles, Production Rejects, and Startup Rejects.

Process Redesign Styles

Transactional Redesign

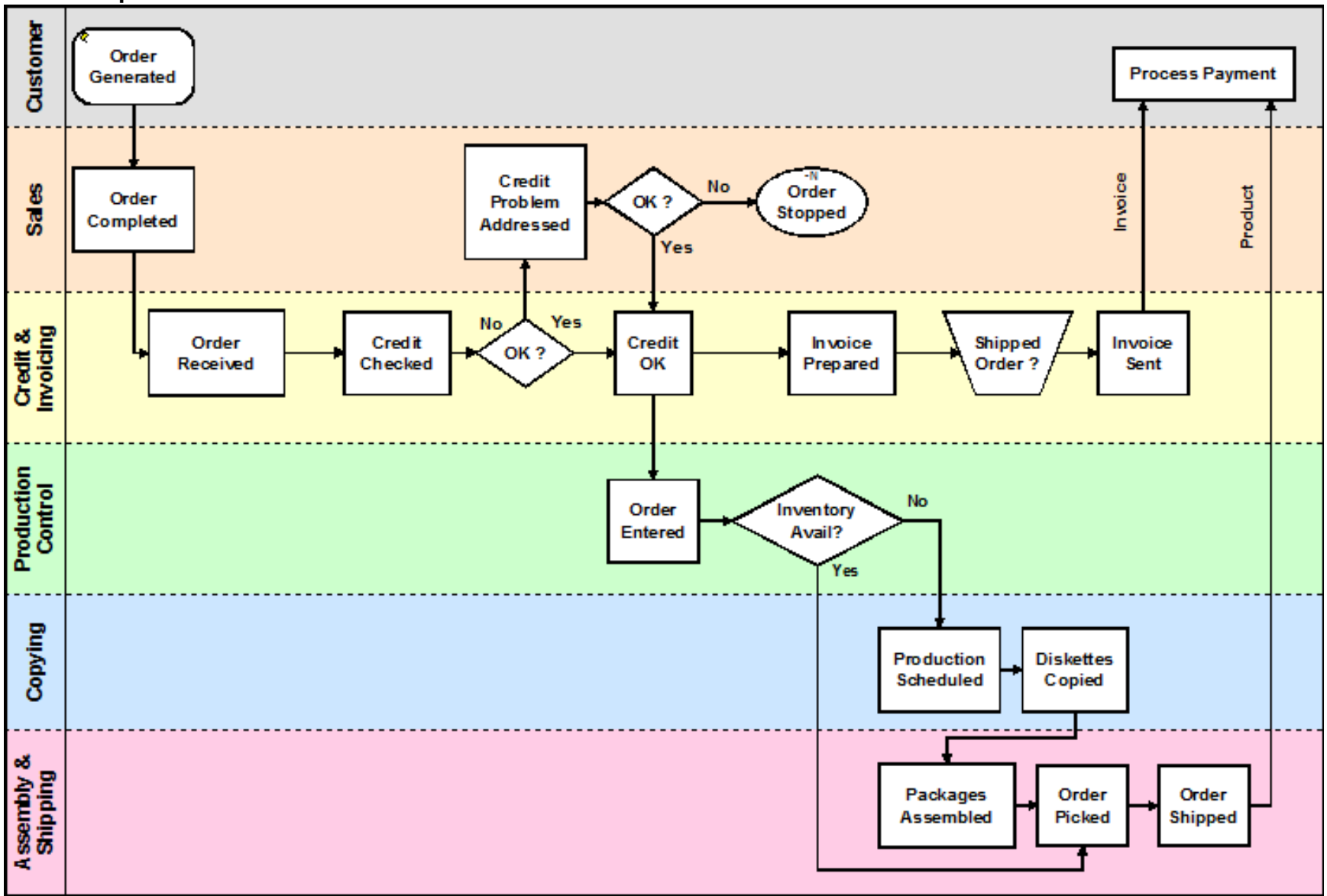
- Doesn't put into question the current process structure
- Seeks to identify problems and resolve them incrementally, one step at a time
- E.g. Business Process Reengineering (BPR)

Transformational Redesign

- Puts into question the fundamental assumptions and principles of the existing process structure
- Aims to radically change the process structure
- E.g. Heuristic process redesign

Process Mapping

- A process map (flowchart) describes the specific steps in a process. The processes could be classified in swim-lanes (colors below).



Analyzing Process Maps

- Are the steps in the process arranged in logical sequence?
- Do all steps add value? Can some steps be eliminated and should others be added in order to improve quality or operational performance? Can some be combined? Should some be reordered?
- Are capacities of each step in balance; that is, do bottlenecks exist for which customers will incur excessive waiting time?
- What skills, equipment, and tools are required at each step of the process? Should some steps be automated?
- At which points in the system might errors occur that would result in customer dissatisfaction, and how might these errors be corrected?
- At which point or points should quality be measured?
- Where interaction with the customer occurs, what procedures and guidelines should employees follow to present a positive image?

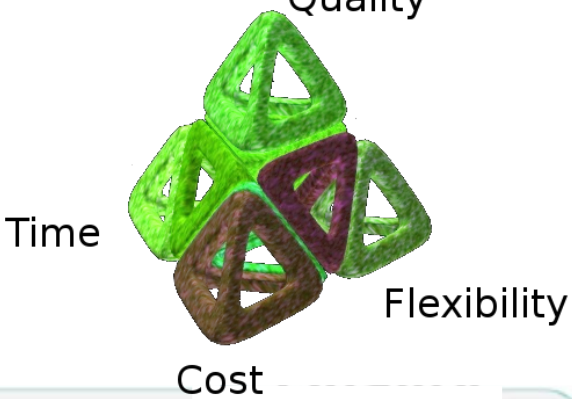
Developing Process Maps – AT&T's backwards approach

1. Begin with the process output and ask, "What is the last essential subprocess that produces the output of the process?"
2. For that subprocess, ask, "What input does it need to produce the process output?" For each input, test its value to ensure that it is required.
3. For each input, identify its source. In many cases, the input will be the output of the previous subprocess. In some cases, the input may come from external suppliers.
4. Continue backward, one subprocess at a time, until each input comes from an external supplier

Improving Process Maps

Using different viewpoints to help with Creative thinking and generating ideas:

Ask if the change will improve quality, save time, make things more cost-effective or flexible?



Task-level

- Task elimination
- Task composition/decomposition
- Triage

Flow-level

- Re-sequencing
- Parallelism enhancement

Process-level

- Specialization & standardization
- Resource optimization
- Communication optimization
- Automation

Mistake-Proofing Processes (poka-yoke)

- **Typical reasons for mistakes and errors:**
  - Forgetfulness due to lack of reinforcement or guidance
  - Misunderstanding or incorrect identification because of the lack of familiarity with process or procedures
  - Lack of experience
  - Absentmindedness and lack of attention, especially when a process is automated

Preventing Mistakes

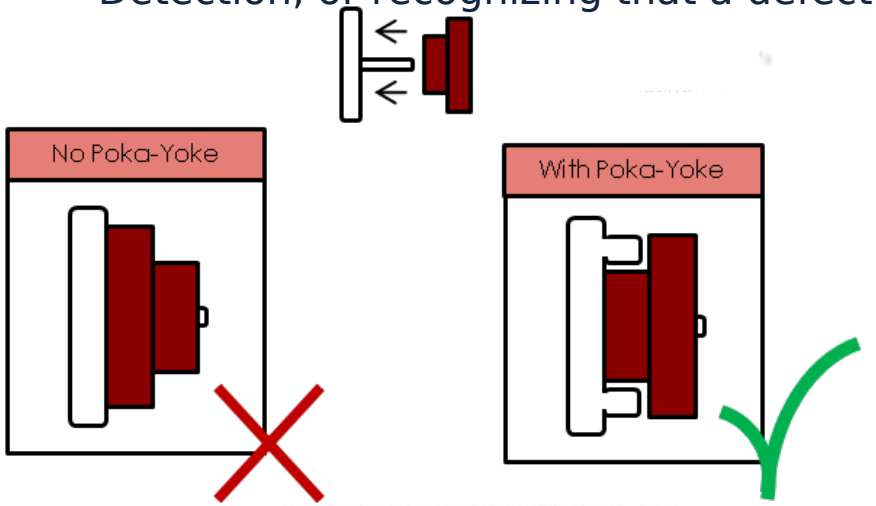
1. Designing potential defects and errors out of the process. This eliminates any possibility that the error or defect will occur and will not result in rework, scrap, or wasted time.
2. Identifying potential defects and errors and stopping a process before they occur. Often results in some non-value-added time.
3. Identifying defects and errors soon after they occur and quickly correcting the process. Results in scrap, rework, and wasted resources.

Types of Service Errors

- Task errors include doing work incorrectly, work not requested, work on the wrong order, or working too slowly.
- Treatment errors in the contact between the server and the customer, such as lack of courteous behaviour, and failure to acknowledge, listen, or react appropriately to the customer.
- Tangible errors, such as unclean facilities, dirty uniforms, inappropriate temperature, and document errors.
- Customer errors in preparation such as the failure to bring necessary materials to the encounter, to understand their role in the service transaction, and to engage the correct service.
- Customer errors during an encounter such as inattention, misunderstanding, or simply a memory lapse, and include failure to remember steps in the process or to follow instructions.
- Customer errors at the resolution stage of a service encounter include failure to signal service inadequacies, to learn from experience, to adjust expectations, and to execute appropriate post-encounter actions.

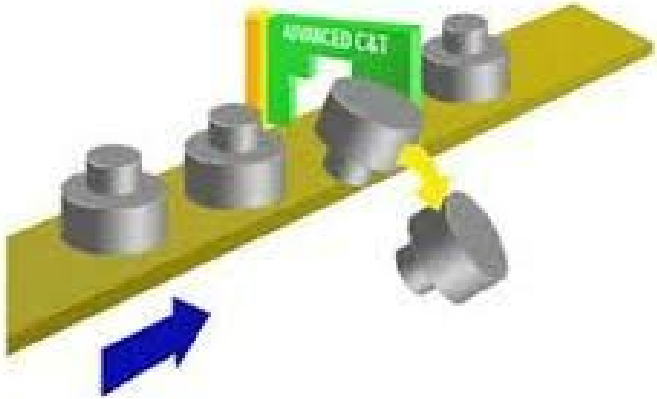
Poka-Yoke

- …an approach for mistake-proofing processes using automatic devices or simple methods to avoid human
- Based on:
  - Prediction, or recognizing that a defect is about to occur and providing a warning
  - Detection, or recognizing that a defect has occurred and stopping the process.



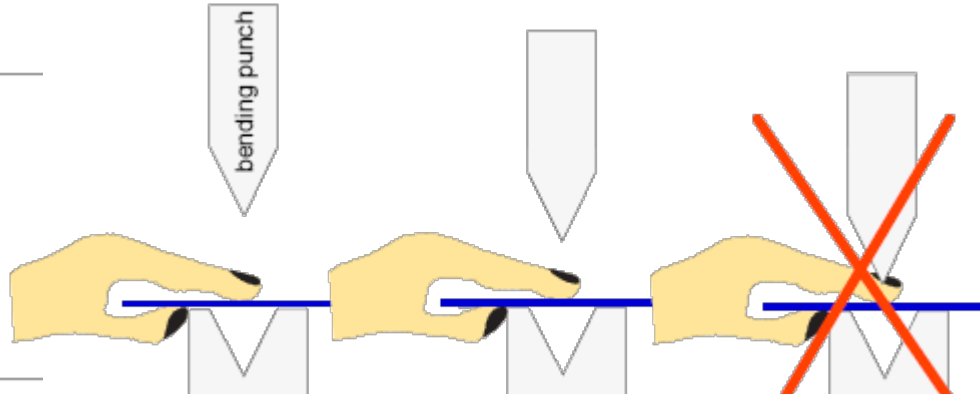
KEY PRINCIPLE

Error is inevitable, but it should be eliminated.



Examples

- Many machines have sensors that would be activated only if the part was placed in the correct position.
- A device on a drill counts the number of holes drilled in a work piece; a buzzer sounds if the work piece is removed before the correct number of holes has been drilled.
- Computer programs display a warning message if a file that has not been saved is to be closed.
- Passwords set for web accounts are entered twice.
- Orders for critical aircraft parts use pre-fit foam forms that only allow the correct part to be placed in them, ensuring that the correct parts are shipped.
- Associates at Amazon sort products into bins that weigh them and compare the weight to the order; if there is an inconsistency, the associate is prompted to verify the items.



bending punch

Agenda

- Process management
- Identifying process and requirements
- Process design
- Process control
- Process improvement
- Managing supply chain processes

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Capture information once and at the source

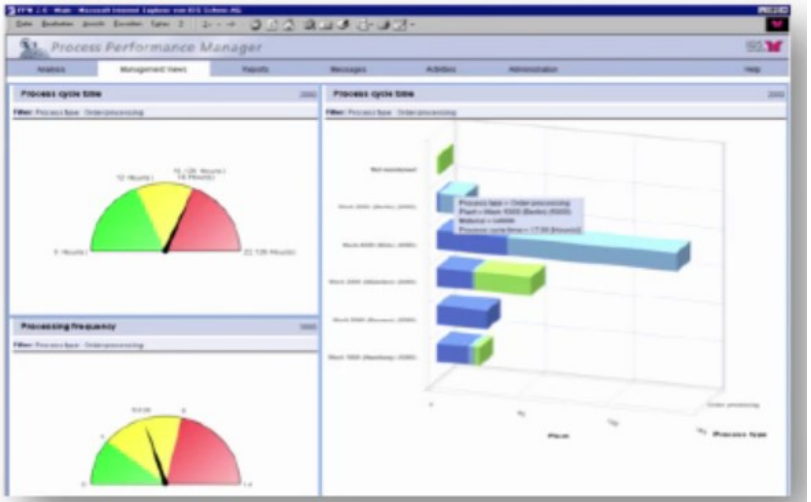
Subsume information-processing work into the real work that produces the information

Have those who use the output of the process drive the process

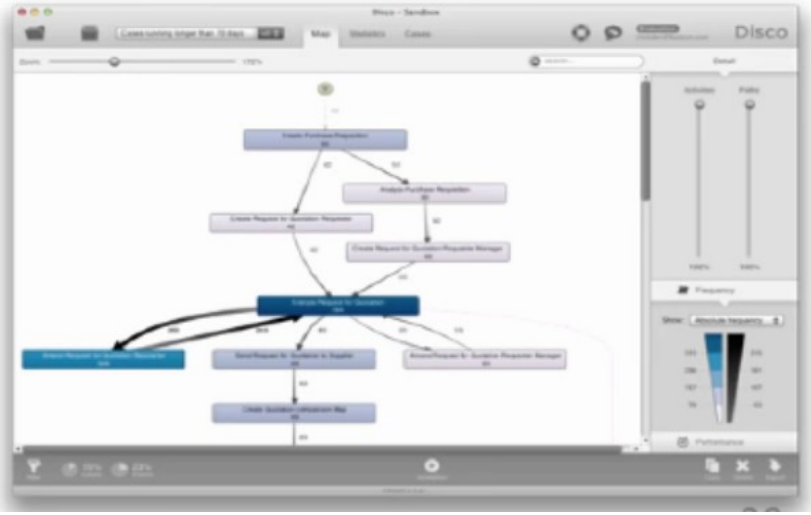
Put the decision point where the work is performed, and build control into the process

Process monitoring

Dashboards & Reports



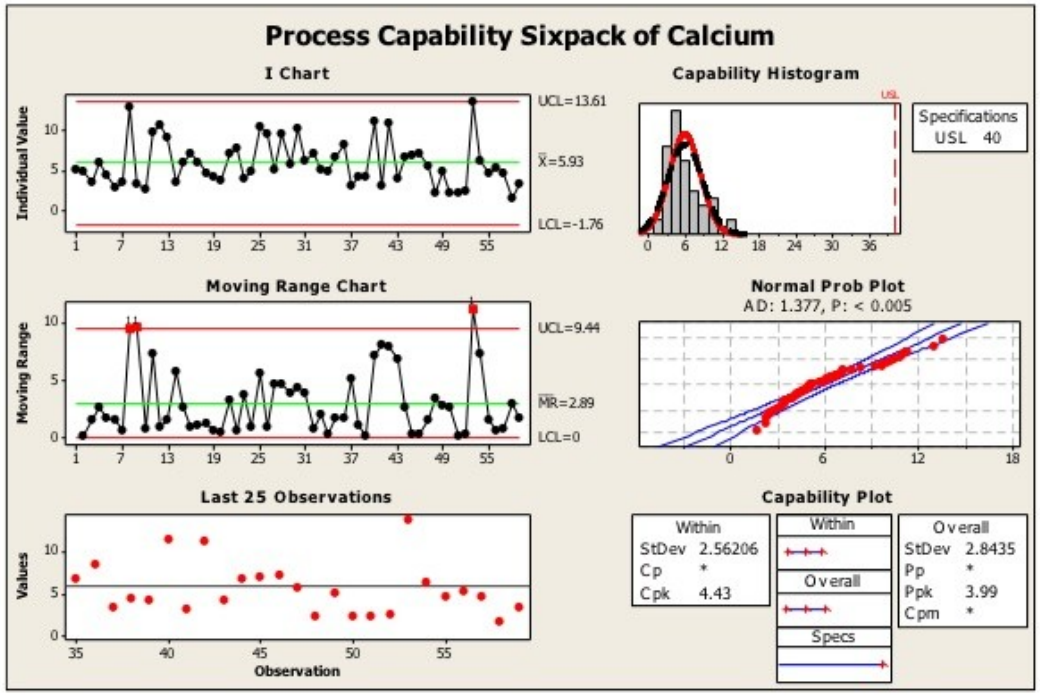
Process Mining



Process Control (covered in detail in later chapters)

- Control – the activity of ensuring conformance to requirements and taking corrective action when necessary to correct problems and maintain stable performance
- Control is different from improvement:

Example of Control Limits and Cpk using a Normal PDF



Control systems

Any control system has four elements:

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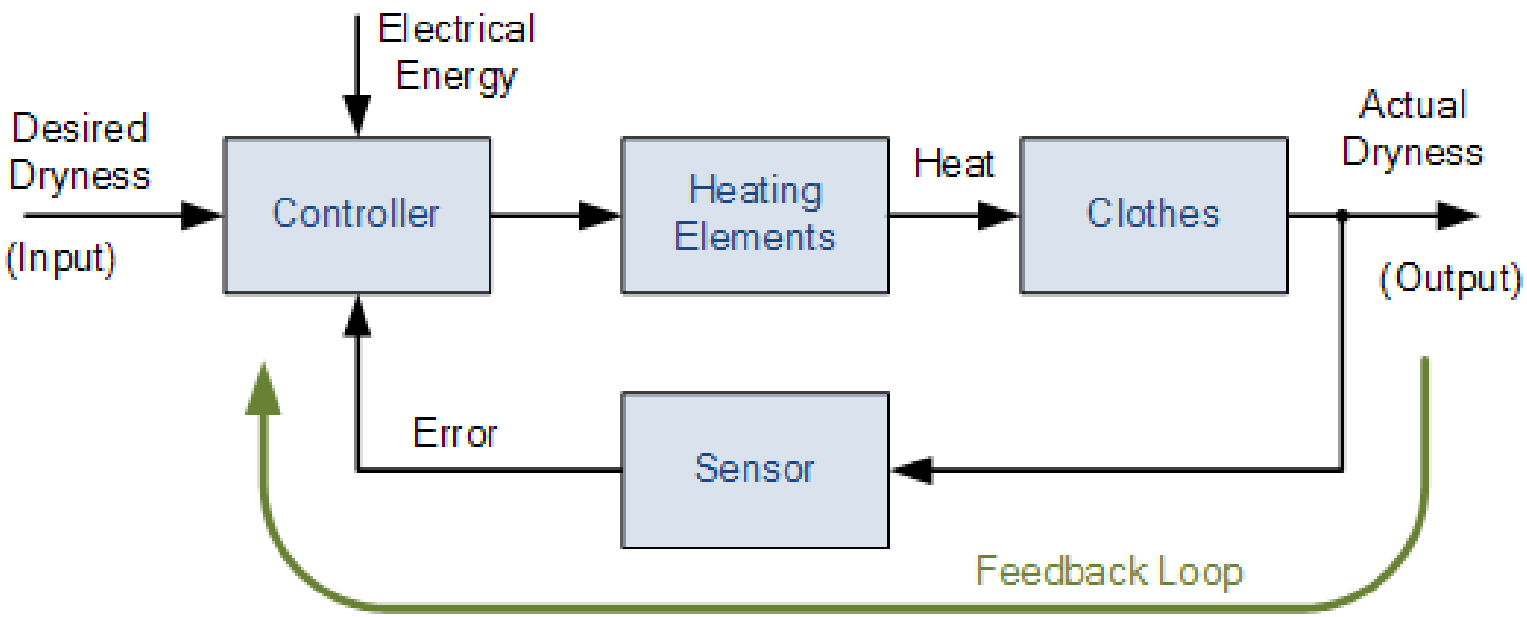

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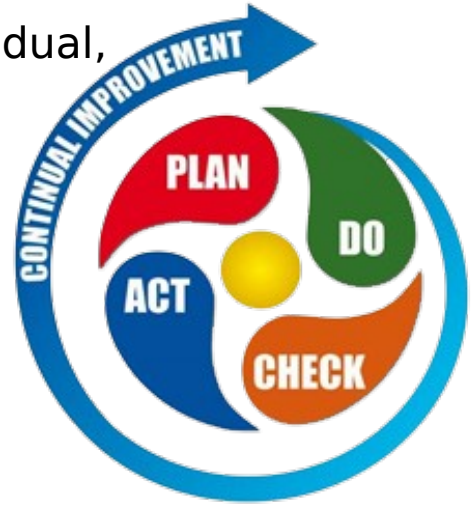
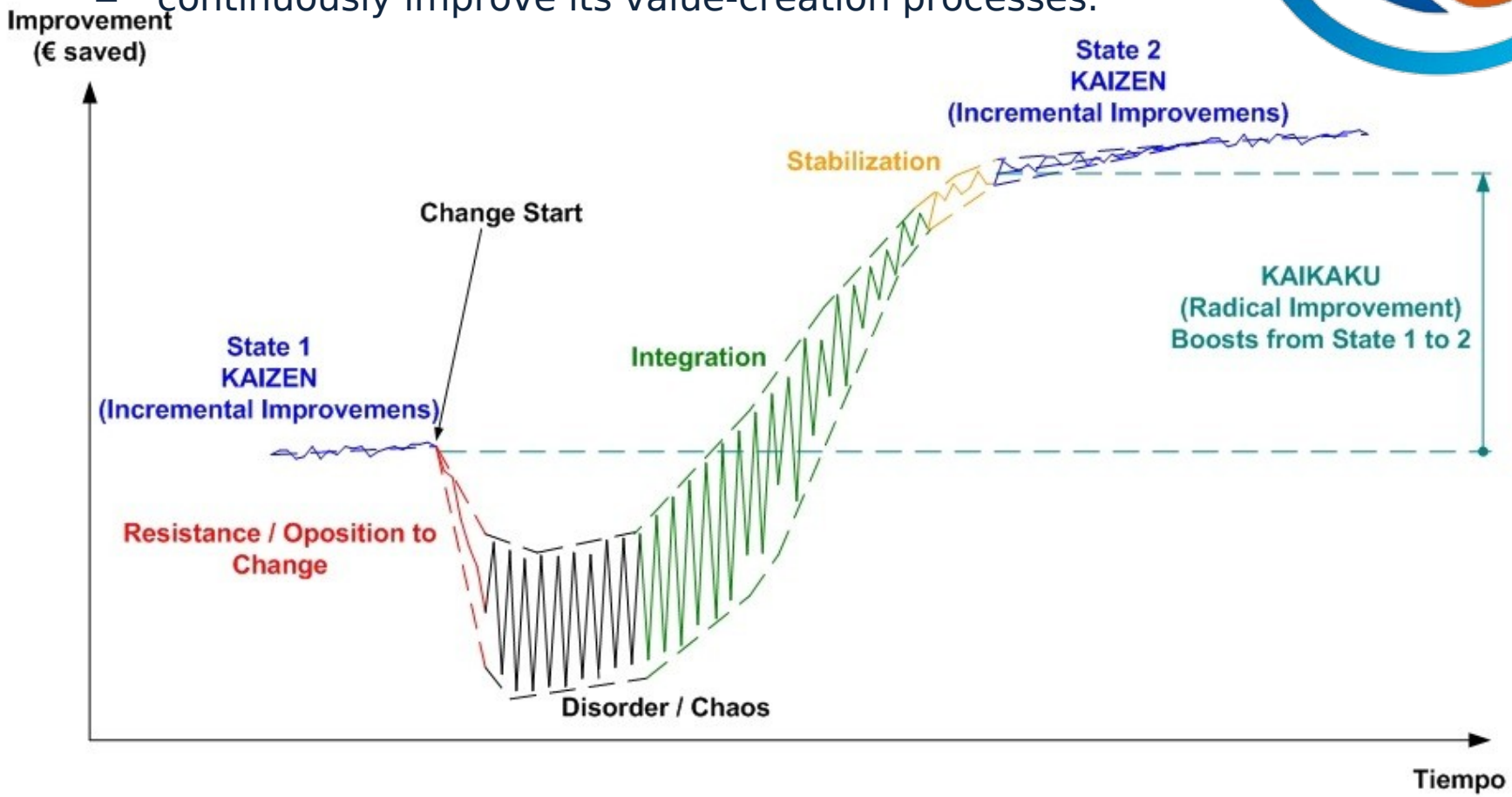
Continuous Improvement (Kaizen)

- …refers to both incremental changes, which are small and gradual, And breakthrough improvements, which are large and rapid.
- Continuous improvement is important because
  - Customer loyalty is driven by delivered value.
  - Delivered value is created by business processes.
  - Sustained success in competitive markets requires a business to continuously improve delivered value.
  - continuously improve its value-creation processes.



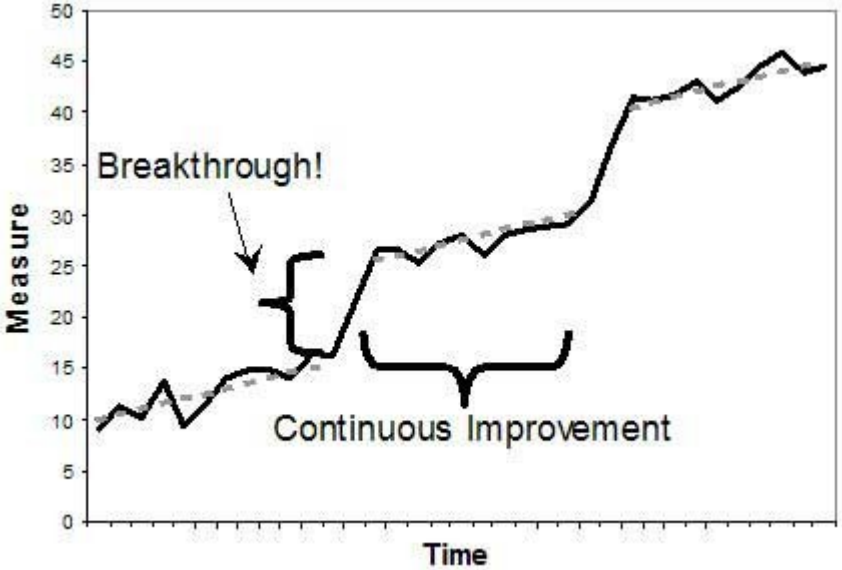


Kaizen

- Kaizen – a Japanese word that means **gradual and orderly** continuous improvement
- Focus on **small, gradual, and frequent improvements** over the long term **with minimum financial investment**, and **participation by everyone in the organization**.

Breakthrough Improvement

- …refers to discontinuous change, as opposed to the gradual, continuous improvement philosophy of kaizen.
- Breakthrough improvements result from innovative and creative thinking; often these are motivated by stretch goals, or breakthrough objectives.
- Example (Motorola):
  - Improve product and services quality ten times within two years, and at least 100-fold within four years



Benchmarking

- Benchmarking – “the search of industry best practices that lead to superior performance.”
- Best practices – approaches that produce exceptional results, are usually innovative in terms of the use of technology or human resources, and are recognized by customers or industry experts.

- **Competitive benchmarking** - studying products or business results against competitors to compare pricing, technical quality, features, and other quality or performance characteristics.
- **Process benchmarking** – identifying the most effective practices in key work processes in organizations that perform similar functions, no matter in what industry.

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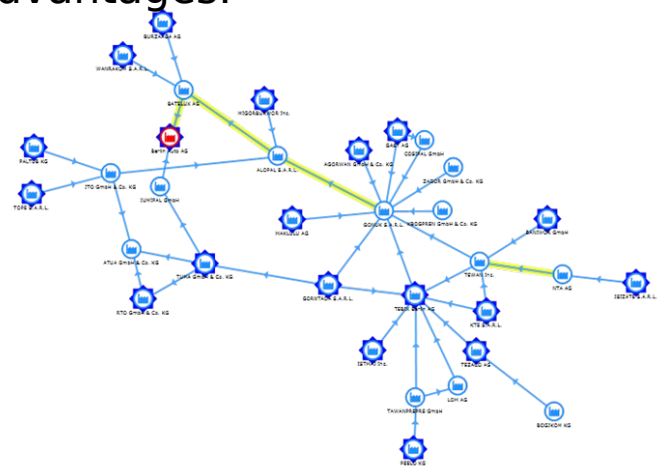
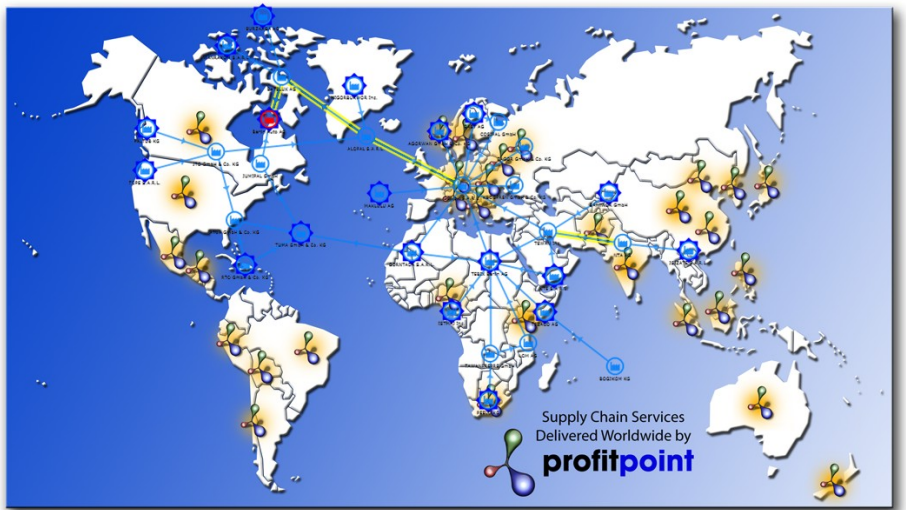
Managing Supply Chain Processes

- Suppliers include not only companies that provide materials and components, but also distributors, transportation companies, and information, health care, and education providers.
- Many companies segment suppliers into categories based on their importance to the business and manage them accordingly.

Principles

- Recognizing the strategic importance of suppliers in accomplishing business objectives, particularly minimizing the total cost of ownership,
- Developing win-win relationships through long-term partnerships rather than as adversaries, and
- Establishing trust through openness and honesty, thus leading to mutual advantages.